Lecture 4
The IS-LM Model

Economics B — Macroeconomics I

Reto Föllmi

University of St. Gallen

Spring 2026

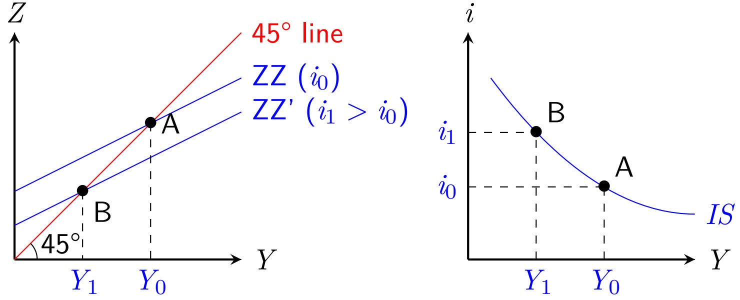
Contents of the Lecture

Learning Objectives and Literature of the 4th Lecture

- Learning objectives:
  - The economy in the short term
  - The IS-LM model
  - Shocks and economic policy
  - The limitations of the IS-LM model
- Literature: Textbook by Blanchard: Chapter 5 & 6

The IS curve – Short repetition

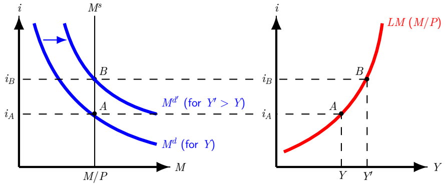
- There is an equilibrium in the goods market if production Y equals the demand for goods Z.
- $Y = C(Y - T) + I(Y, i) + G$ (assumption: $EX = IM = 0$)
- As the interest rate rises, income decreases in the goods market equilibrium. The IS curve therefore has a downward slope.

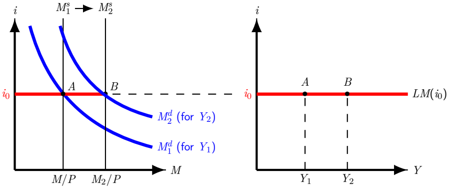


The LM Curve — Short repetition

- The financial market is in equilibrium when the real money supply is equal to the real money demand: $M/P = Y * L(i)$

- We distinguish between money supply control (left-hand chart) and interest rate control (right-hand chart):



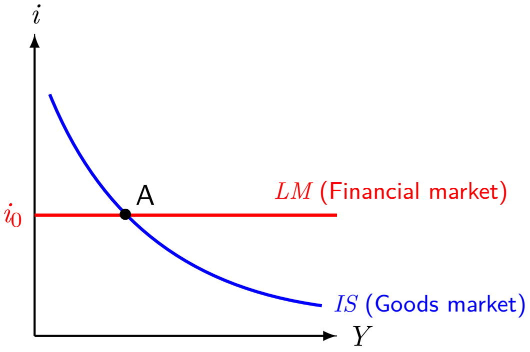


IS curve and LM curve together: IS-LM Model

- John Hicks and Alvin Hansen developed the IS-LM model in the late 1930s and early 1940s.
  - Their approach was based in large part on John Maynard Keynes' book "General Theory of Employment, Interest, and Money" (1936).
- While in the original model the central bank controls the money supply, we usually consider the case of interest rate control.
- IS curve: $Y = C(Y - T) + I(Y, i) + G$
- LM curve: $i = i_0$

IS curve and LM curve together: IS-LM Model (Graph)

- Only in point A, the intersection of both curves, we have a simultaneous equilibrium in the goods and financial markets.
- Graphical representation of the IS-LM model:



Comparative statics in IS-LM model

- When conducting comparative statics, we investigate how the equilibrium levels (of i and Y) adjust as we change one variable.
  - We change exogenous variables such as T, G, or M^s.
- We have to distinguish between factors that cause a movement along the IS or LM curve and those that cause a shift of the curve.
  - Factors that trigger a decrease (an increase) in the demand for goods at a given interest rate shift the IS curve to the left (right).
    Example: government spending.
  - An increase in money supply or a reduction in the minimum reserve requirements shift the LM curve downwards.

Fiscal policy

- State influence on the economy: Fiscal and monetary policy
- Fiscal policy:
  - Tax reduction / increase: e.g., income tax, value added tax
  - Higher / lower transfer payments: e.g., unemployment benefit
  - Higher / lower government spending: e.g., infrastructure, education, military
- Budget of the state is determined by G and T (budget deficit if $G > T$)
- Expansionary fiscal policy: $G \uparrow$ and/or $T \downarrow$
- Contractionary fiscal policy: $G \downarrow$ and/or $T \uparrow$
- Changes in G and T affect the IS curve, not the LM curve.

Expansionary fiscal policy in the IS-LM model

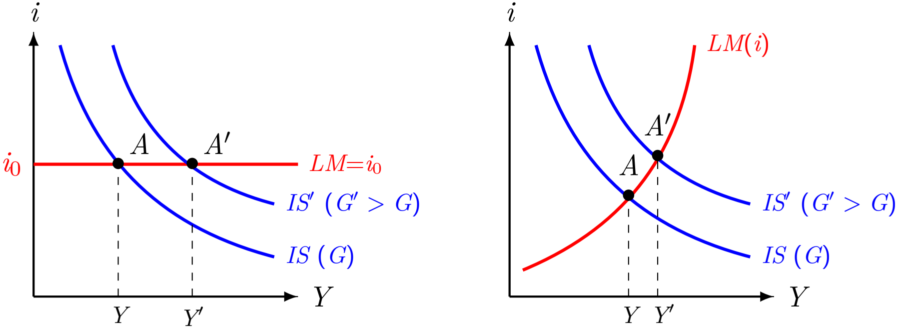
- Example 1 (assumption: underutilized capacity): The state increases G, for example by building new roads.
- Transmission mechanism:
  - $G \uparrow \rightarrow$ demand for goods $Z \uparrow \rightarrow$ production $\uparrow \rightarrow$ income $Y \uparrow \rightarrow$ *multiplier effect*: [if $Y \uparrow$, then C and $I \uparrow$, so Y continues to rise] $\rightarrow$ $Y \uparrow \rightarrow$ demand for money $(PY * L(i)) \uparrow \rightarrow$ money supply $(M_S) \uparrow \rightarrow$ i remains unchanged
- The interest rate i remains unchanged with interest rate control by the central bank.
  With money supply control (increasing LM curve) i rises as a result of higher government spending.
- As a result, expansionary fiscal policy has increased GDP.

Expansionary fiscal policy in the IS-LM model (Graph)

- Expansionary fiscal policy shifts the IS curve to the right.
- The GDP (Y) increases by more than ΔG ($\rightarrow$ multiplier effect)
- If the central bank controls the money supply, the interest rate i rises and the positive effect on GDP (ΔY) is smaller than with interest rate control.
  - Government spending then crowds out private investment.
  - Why? Private investment depends negatively on i: $I = I(\underset{(+)}{Y}, \underset{(-)}{i})$
    - The central bank keeps money supply fixed, but money demand increases, so i in equilibrium is rising in the money market.

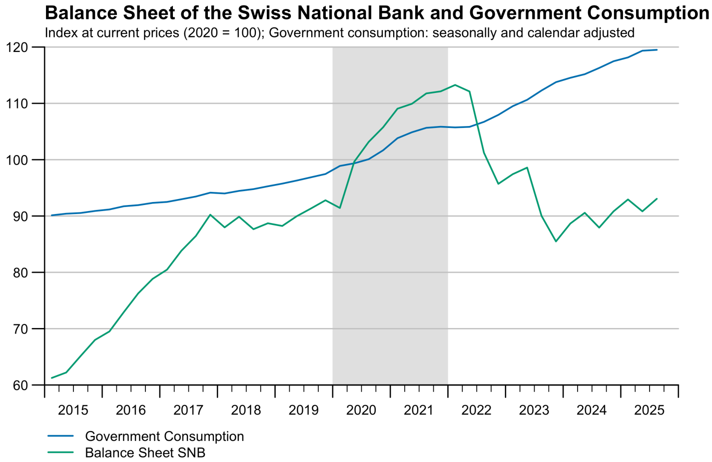
Expansionary Fiscal Policy in the IS-LM Model (Graph)

- Graphical representation: Interest rate control vs. money supply control



Expansionary Fiscal Policy in the IS-LM Model (Practical Example)

- The corona crisis provides a good example of expansionary fiscal policy:

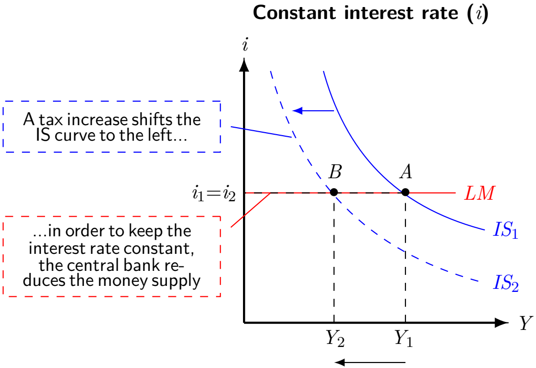


Contractionary fiscal policy in the IS-LM model

- Example 2: The state raises taxes T:
- Transmission mechanism (for interest rate control):
  - $T\uparrow \rightarrow Y_D\downarrow \rightarrow C\downarrow \rightarrow$ demand for goods $\downarrow \rightarrow$ Production $\downarrow \rightarrow Y\downarrow \rightarrow$ *multiplier effect* [if $Y\downarrow$, then $C\downarrow$ and $I\downarrow$, thus Y further $\downarrow$] $\rightarrow Y\downarrow\rightarrow$ Demand for money $(PY * L(i))\downarrow \rightarrow$ Money supply $(M_S)\downarrow \rightarrow i$ unchanged
- Transmission mechanism (for money supply control):
  - $T\uparrow \rightarrow Y_D\downarrow \rightarrow C\downarrow \rightarrow$ demand for goods $\downarrow \rightarrow$ Production $\downarrow \rightarrow Y\downarrow \rightarrow$ *multiplier effect* [if $Y\downarrow$, then $C\downarrow$ and $I\downarrow$, thus Y further $\downarrow$] $\rightarrow Y\downarrow\rightarrow$ Demand for money $(PY * L(i))\downarrow \rightarrow$ Money supply (M_S) constant $\rightarrow i\downarrow \rightarrow I\uparrow$
- The decline in GDP due to contractionary fiscal policy is smaller, if the central bank lets the interest rate fall.

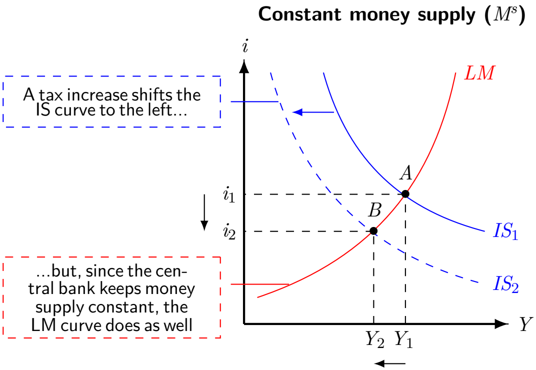
Contractionary fiscal policy in the IS-LM Model (Graph I)

- Effect of a tax increase depends on how the central bank acts.
- Case 1 — Interest rate control: Central bank keeps interest rate constant



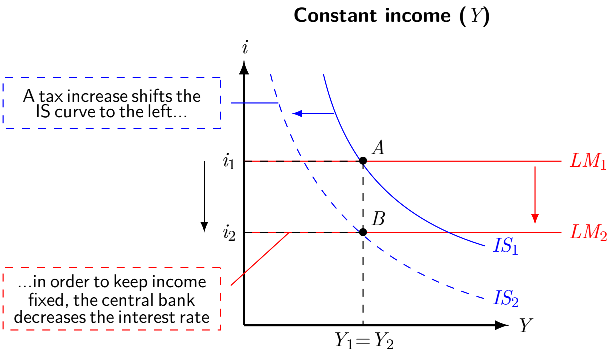
Contractionary fiscal policy in the IS-LM Model (Graph II)

- Case 2 — Money supply control: Central bank keeps money supply constant



Contractionary Fiscal Policy in the IS-LM Model (Graph III)

- Case 3 — Income control: Central bank keeps Y constant



Fiscal policy — Discussion

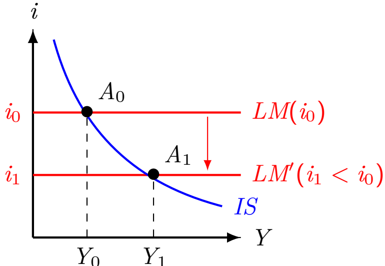
- The effects of fiscal policy also depend on what monetary policy the central bank is pursuing.
- With underutilized capacity, expansionary fiscal policy can raise Y. However, it should be noted:
  - Crowding out of private investment
  - Value-for-money: Efficiency of government spending
  - Time lag: Effect is time-delayed
  - Leakages: Multiplier reduced by savings and imports
  - Ricardian equivalence: People expect future tax increase
- In the medium run, state budget restrictions limit the scope of expansionary fiscal policy.
- $\longrightarrow$ further discussion in public finance (*public economics*)

Monetary Policy in IS-LM Model

- Expansionary monetary policy:
  A reduction in the interest rate or an increase in the money supply shift the LM curve downwards.
- Contractionary monetary policy:
  An increase in the interest rate or a reduction in the money supply shift the LM curve upwards.
- Monetary policy does not shift the IS curve.

Expansionary monetary policy in the IS-LM model

- Example: In March 2020, the US Federal Reserve lowered interest rates
- Transmission mechanism:
  - $M^S \uparrow \to i \downarrow \to I(i) \uparrow \to$ demand for goods $\uparrow \to Y$ (production and income) $\uparrow$ $\to$ [...multiplier effect...] $Y \uparrow \to$ Demand for money $(PY * L(i)) \uparrow \to$ Money supply $(M_S) \uparrow$
    (Central bank adjusts the money supply to meet the increased money demand).

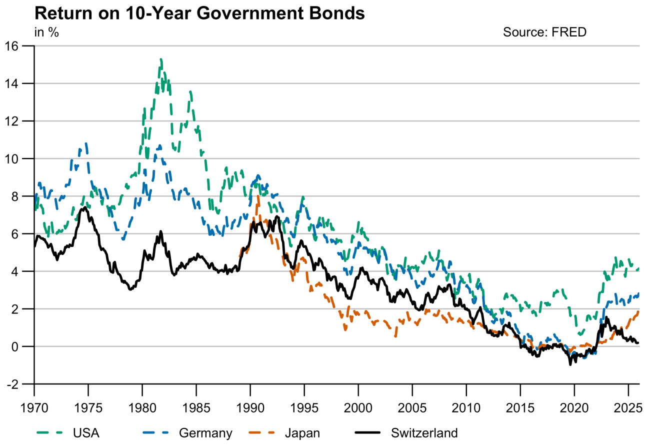


Monetary policy discussion

- Monetary policy has always been controversial:
  - "Not all Germans believe in God, but they all believe in the Bundesbank." — Jacques Delors (President of the European Commission, 1985–1995)
- Debates:
  - Does loose monetary policy generate inflation? ($\longrightarrow$ lecture 6)
  - Liquidity trap $\longrightarrow$ next three slides
  - Credibility of the central bank (e.g. *Greenspan put*, moral hazard)
  - Effects on the exchange rate ($\longrightarrow$ lectures 8 and 9)
  - SNB: room for maneuver on interest rates exhausted after the financial crisis, hence accumulation of foreign exchange reserves (purchase of foreign assets) to reduce appreciation pressure. Discussion about distribution of resulting profits. (NZZ article)

Limits of monetary policy — during the Corona crisis

- Very low interest rates since the financial crisis:



Limits of monetary policy — during the Corona crisis

- With interest rates very low around the world, warnings of a liquidity trap were issued in January 2020:

Central banks low on ammunition to fight recessions, warns Carney

- Outgoing BoE chief highlights liquidity trap
- Call to exploit low-carbon opportunities

LIONEL BARBER AND CHRIS GILES — LONDON

The global economy is heading towards a "liquidity trap" that will undermine central banks' efforts to avoid a future recession, according to Mark Carney, governor of the Bank of England.

A liquidity trap occurs in the rare occasions when monetary policy loses all effectiveness to manage economic swings and looser policy does not encourage any additional spending.

In a wide-ranging interview with the Financial Times, the outgoing governor warned that central banks were running out of the ammunition needed to combat a recession.

"It's generally true that there's much less ammunition for all the major central banks than they previously had and I'm of the opinion that this situation will persist for some time," he said.

That meant there was a need to look for supplements to monetary tools, including interest rate cuts, quantitative easing and guidance on future interest rates, he said. "If there were to be a deeper downturn, [that requires] more stimulus than a conventional recession, then it's not clear that monetary policy would have sufficient space."

Mr Carney echoed other central bankers, such as the European Central Bank's Mario Draghi and his successor, Christine Lagarde, in recommending that governments consider fiscal policy tools, such as tax cuts or public spending increases when tackling a downturn. However, he accepted "it's not [central bankers'] job to do fiscal policy".

The governor said monetary policy was not yet a spent force internationally, with US and eurozone interest rate cuts last year encouraging borrowing and spending. "We're starting to see that stimulus flow to the global economy."

He insisted that he was not leaving his successor, Andrew Bailey, without any tools in his armoury. The BoE could still cut interest rates from 0.75 per cent to close to zero and "supplement monetary policy with macroprudential tools" by relaxing banks' capital requirements to enable them to lend more.

Mr Carney predicted that the City of London could profit from the "huge commercial opportunity" of helping to finance and accelerate action to mitigate global warming — although he recognised that the financial sector was not a substitute for effective policies at the national and international level.

Mark Carney: 'We have a statutory responsibility to identify the major risks to financial stability. We can't dodge that'

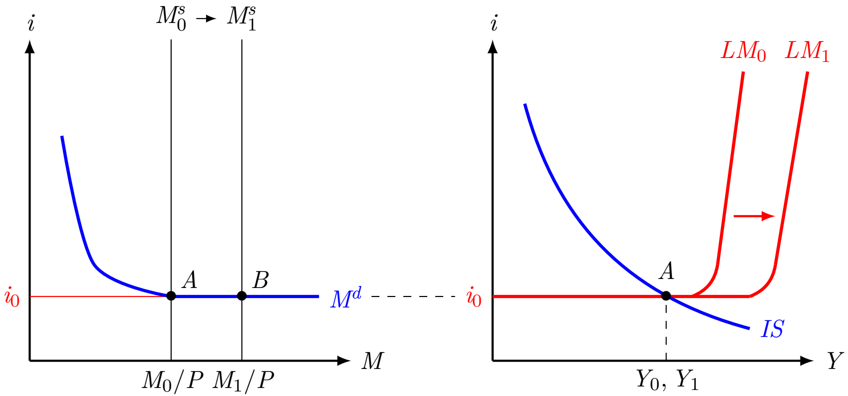
He predicted that the City could benefit from financing the transition to a low-carbon economy in place of some EU activity. "This happens to be a huge commercial opportunity for the City of London and the UK financial sector writ large."

The Bank of England has led other central banks on taking a firmer stance on combating the financial risks to banks and insurance companies that stem from global warming.

Mr Carney has been criticised for straying beyond his mandate with this position but he dismissed those concerns. "Anybody who says that doesn't know the breadth of the powers of the Bank of England," the governor said.

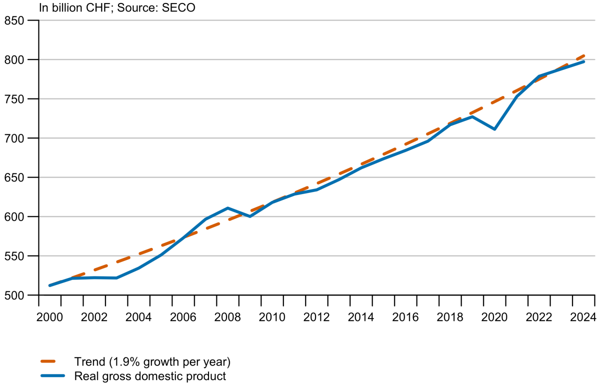
Source: *Financial Times*, front page, January 8, 2020

The Liquidity Trap (Graph)



Business cycles

- The goods market (the GDP) is subject to fluctuations over time.
- Example: Switzerland's GDP since 2000



Business cycles — Explanatory approaches

- GDP: $Y = C + I + G + NX$
  - Fluctuations in GDP can have many causes.
  - Many factors determine C, I, G, NX.
- Impulse/shock $\longrightarrow$ transmission $\longrightarrow$ interaction $\longrightarrow$ economic fluctuation
- Possible stimuli: Fiscal & monetary policy measures, technological leaps, demand shocks, financial crises (Minsky hypothesis)
- The IS-LM model helps with:
  - Analysis of business cycles
  - Forecast of short-run GDP development (and its difficulty!)

Business cycles — Comparison across industries

- Not all industries are affected by the business cycle to the same extent.
- A positive correlation with the business cycle ($Corr > 0$) indicates that a sector is pro-cyclical.

Selected correlations between sector value added growth and GDP growth

Sector	$Corr$
Manufacturing (Noga 10-33)	0.70
Business services (Noga 68-75, 77-82)	0.57
Hospitality (Noga 55-56)	0.42
Retail trade (Noga 47)	0.39
Health and social services (Noga 86-88)	0.15
Education (Noga 85)	0.05
Construction (Noga 41-43)	0.05
Water, sewage, waste (Noga 36-39)	-0.42

Real, seasonally, calendar and sports-event adjusted, 1990 - 2024, excl. Covid years 2020 & 2021. Data: SECO

Extensions of the IS-LM Model

- The IS-LM model can be extended in many ways.
  - Difference between nominal and real interest rates, risk premiums $\longrightarrow$ Lecture 6 and chapter 6 in the textbook
  - Consumption: $C = C(Y_D)$ may also depend on expectations (*consumer confidence*) or assets (IS-LM model ignores *stocks*)
  - Investments: may depend on expectations ($\longrightarrow$ PMI, investment trap)
  - Open economy: exports and imports, exchange rates
- Important three steps:
  Understanding model, applying model, questioning model.

Extensions of the IS-LM Model (continued)

- Calvin & Hobbes meet Keynesian politics:

Source: Calvin and Hobbes, 1992, Watterson, Universal Press Syndicate

Limitations of the IS-LM model

- The model provides an initial framework for economic analysis.
  In this course as well as in the further study of economics,
  central questions will be examined in more detail:
  - How do the short- and medium-run effects of expansionary or contractionary fiscal policy differ?
  - Should the state better reduce taxes ($T\downarrow$) or increase spending ($G\uparrow$)?
  - What else can the central bank do when the interest rate is already at zero?
  - Are the incentives for key decision-makers in fiscal and monetary policy identical to those of the population? ($\longrightarrow$ *political economy*)

Explanatory power of the IS-LM model

- In god we trust, others must provide data. — Edwin R. Fisher
- Do economic data (or an econometric analysis) support the core statements of the IS-LM model?
  - Various empirical studies support the core statements of the model and also show the dynamics (time lag).
    $\longrightarrow$ chapter 5.5 in the textbook
  - Difficulty for empirical studies:
    Large, isolated shocks and policy changes are rare.

Summary and Outlook

- We now have an economic model for the analysis of the economy in the short run: the IS-LM model.
- Like any economic model, the IS-LM model has limitations, but it can be expanded.
- In the model, G and thus Y can be increased at will. What is the limit?
- In the medium-run, resource constraints are important. Furthermore, prices are not stable and inflation can occur — for example due to expansionary monetary policy.
- In lectures 5 and 6 we will therefore examine the labor market and inflation.
- Preparation for lecture 5: Slides and chapter 7 in the textbook